Day 8 — Configure GoDaddy Domain with Route 53 + ALB + SSL

Objective

Today we configured a custom domain purchased from GoDaddy and connected it to an AWS Application Load Balancer (ALB) using Route 53.

We also configured SSL/HTTPS using AWS Certificate Manager (ACM).

Architecture

User

↓

saikumar.online

↓

Route 53

↓

Application Load Balancer (ALB)

↓

Target Group

↓

EC2 Instance / Auto Scaling

↓

NGINX Application

Step 1 — Verify Load Balancer

We already had an ALB created in AWS.

Example:

http://nginx-1317210205.us-east-1.elb.amazonaws.com

This confirmed:

- ALB was working
- Target Group was attached
- EC2 instances were serving traffic

Step 2 — Create Route 53 Hosted Zone

Opened:

AWS Console → Route 53 → Hosted Zones

Created a Public Hosted Zone for:

saikumar.online

AWS automatically generated:

- NS Record
- SOA Record

Step 3 — Copy Route 53 Nameservers

Example nameservers:

ns-123.awsdns-45.com

ns-456.awsdns-78.net

ns-789.awsdns-12.org

ns-321.awsdns-99.co.uk

These nameservers tell the internet that AWS Route 53 will manage DNS.

Step 4 — Update GoDaddy Nameservers

Opened:

GoDaddy → Domain Settings → Nameservers

Selected:

Use my own nameservers

Added all 4 Route 53 nameservers.

Saved changes.

Step 5 — Wait for DNS Propagation

DNS propagation time:

- Usually 15–60 minutes
- Sometimes up to 24 hours

Verified using:

nslookup saikumar.online

and

DNSChecker.org

Step 6 — Create Route 53 Alias Record

Opened:

Route 53 → Hosted Zone → Create Record

Configured:

Setting	Value
Record Type	A
Alias	ON
Target	Application Load Balancer

Selected ALB:

dualstack.Nginx-1317210205.us-east-1.elb.amazonaws.com

Important:

Do NOT use ALB IP addresses because they change dynamically.

Correct method:

Alias A Record → ALB

Step 7 — Verify Domain Access

After DNS propagation:

http://saikumar.online

started opening successfully.

This confirmed:

- Route 53 configuration was correct
- ALB routing was correct
- Target Group was healthy
- EC2 + NGINX were working

Step 8 — "Not Secure" Issue

Browser showed:

Not Secure

Reason:

Site was using:

HTTP (Port 80)

instead of:

HTTPS (Port 443)

To fix this, SSL certificate setup was required.

Step 9 — Configure AWS Certificate Manager (ACM)

Opened:

AWS Console → Certificate Manager (ACM)

IMPORTANT:

Certificate must be created in SAME REGION as ALB.

Region used:

us-east-1

Step 10 — Request Public SSL Certificate

Requested certificate for:

saikumar.online

and

www.saikumar.online

Validation method selected:

DNS Validation

Step 11 — ACM Validation

ACM created DNS validation CNAME records.

Clicked:

Create records in Route 53

Certificate status initially:

Pending validation

After DNS validation:

Issued

Typical time:

- 2–10 minutes
- Sometimes 30 minutes

Step 12 — Configure HTTPS Listener on ALB

Opened:

EC2 → Load Balancers → Listeners

Added listener:

Protocol	Port
HTTPS	443

Attached ACM certificate.

Configured forwarding:

HTTPS → Target Group

Step 13 — Redirect HTTP to HTTPS

Edited existing HTTP listener.

Configured redirect:

HTTP:80 → HTTPS:443

Status code used:

HTTP_301

Step 14 — Security Group Configuration

ALB Security Group

Allowed inbound:

Type	Port
HTTP	80
HTTPS	443

Source:

0.0.0.0/0

EC2 Security Group

Allowed inbound:

Type	Port	Source
HTTP	80	ALB Security Group

Step 15 — Final Result

Successfully configured:

- Custom domain
- Route 53 DNS
- ALB integration
- HTTPS SSL
- ACM certificate
- HTTP → HTTPS redirect

Final secure URL:

https://saikumar.online

Browser now shows:

Secure 🔒

Important Learnings

1. ALB Uses Dynamic IPs

Never use ALB IP addresses directly.

Always use:

Alias Record → ALB DNS

---

2. Route 53 Is Best for AWS

Benefits:

- Easy ALB integration
- Easy SSL validation
- Better DNS management
- Alias records supported

---

3. ACM Certificates Are Free

AWS ACM provides free SSL certificates.

Best practice:

- Use ACM with ALB
- Avoid Certbot inside EC2 when using ALB

---

4. SSL Certificate Must Match ALB Region

If ALB region:

us-east-1

Then ACM certificate must also be:

us-east-1

Otherwise certificate will not appear in ALB dropdown.

Commands Used

Check NGINX

sudo systemctl status nginx

Start NGINX

sudo systemctl start nginx

Enable NGINX

sudo systemctl enable nginx

Test Local Server

curl localhost

DNS Lookup

nslookup saikumar.online

Final Production Architecture

Internet User

↓

Custom Domain

↓

Route 53

↓

Application Load Balancer

↓

Auto Scaling Group

↓

EC2 Instances

↓

NGINX / Application

Day 8 Summary

Today we learned:

- Route 53 Hosted Zone creation

- GoDaddy nameserver configuration
- ALB DNS mapping
- Alias Record setup
- ACM SSL certificate configuration
- HTTPS listener setup
- HTTP to HTTPS redirection
- Security group configuration
- Production-ready AWS architecture

This setup is commonly used in real-world production AWS environments.